SHABAZ AHMAD HASIB

Machine Learning Engineer

Hyderabad, India · shabazahmad.me1@gmail.com
linkedin.com/in/shabaz-ahmad-8a4941366 · github.com/shabazahmadme1-eng

EXPERIENCE

Software Engineering Intern — Purser | purserflow.com **Aug 2026 – Present**

WhatsApp guest operations for hotels · TypeScript, Fastify, Supabase Postgres, Groq, WhatsApp Cloud API, Cloud Run, Cloudflare

- Top contributor to the codebase: **224 merged pull requests**, two thirds of everything merged, taking the product from an early prototype to a **live hotel pilot** in 10 weeks.
- Built guest-facing modules end to end: room QR-code check-in, food ordering, itemised guest bills with optional GST invoices, and **Razorpay payments** with webhooks, link-expiry sweeps and English/Hindi copy.
- Added voice-note understanding (transcripts for Urdu and Indian-script messages), routing for messages that carry several requests at once (**110/110** on real-model checks), and an owner mode that answers questions about the hotel on WhatsApp.
- Built the reliability layer: per-message traces with run replay, five chaos modes for broken model output, durable inbound recovery, rate limits, CI safety gates, and **production alerting** with a probe and SLOs.

Machine Learning Research Intern — Neuroparadigm **Feb 2026 – Present**

- Autism screening from gait: fused a two-stream MS-G3D graph network with **534 handcrafted biomechanical features** on 100 children (50 ASD / 50 TD); child-grouped 5-fold CV over 3 seeds gave **93.3% ± 1.2 accuracy** and **0.979 ± 0.007 AUC**.
- Distilled an Explainable Boosting Machine surrogate (in-sample fidelity **r = 0.96**) so clinicians can read each prediction; built a cross-device phone prototype on bone-direction inputs with Platt calibration.
- Webcam eye-tracking: engineered **33 calibration-invariant gaze features** and an 18-biomarker panel; TabICL reached **0.919 AUC** in-domain and **0.799** when moved from lab eye-trackers to webcams.
- Shipped React, Next.js and Tailwind modules for the Neuroparadigm and K-Hub research platforms.

Machine Learning & Computer Vision Intern — Sequetrics **Aug 2025 – Present**

- Built a **12-class YOLOv8** detector for runway cracks, spalling and foreign object debris (**0.693 mAP50**), trained on RDD2022 plus proprietary drone imagery labelled through pseudo-labelling and OpenCV colour-based auto-labelling.
- Optimised the model with **ONNX and TensorRT** and deployed it on **NVIDIA Jetson** for real-time on-device inference.

RESEARCH & PATENTS

- **Interpretable fusion of learned and biomechanical skeletal gait representations for autism-related motor atypicality** — co-author, submitted to *Neuroscience Informatics*, 2026.
- **Calibration-Invariant Webcam Eye-Tracking for Autism Screening** — co-author, under review.
- **Indian Patent App. 202641080489 A** — Browser-Based Real-Time Multimodal Skeletal Data Acquisition with Adaptive Quality Gating (co-inventor).
- **Indian Patent App. 202641079295 A** — Privacy-Preserving Child-Adaptive Webcam Eye-Tracking and Structured Gaze Data Export (co-inventor).

PROJECTS

MIPO-NDD — protein mutation-effect model (in progress) **Python, PyTorch, ESM2**

- ~1.05M-parameter mutation-response model over frozen ESM2-650M features; built and audited a **1.17M-measurement, 501-gene** corpus with gene-held-out splits. Early transfer beats a zero-shot ESM baseline; evaluation ongoing.

ResQbox — food-rescue marketplace backend **TypeScript, Fastify, SQLite**

- Went past a screening brief to build a real backend: transactional orders that cannot oversell, idempotent order creation, signed and deduplicated payment webhooks, an append-only audit log, and **73 automated tests**.

QuantCrate — quantitative research platform **Python, React, Node.js**

- Implemented logistic regression, MLPs and gradient-boosted trees **from scratch** (no ML libraries) on four factor families over 12 years of market data, with causal splits, purge gaps and walk-forward backtests.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL
ML & research: PyTorch, GNNs (MS-G3D, ST-GCN), XGBoost, LightGBM, CatBoost, scikit-learn, EBM, TabICL, ESM2
LLM & backend: Groq, structured JSON outputs, Node.js, Fastify, PostgreSQL, Supabase, Drizzle, WhatsApp Cloud API
Vision & edge: YOLOv8, OpenCV, MediaPipe, ONNX, TensorRT, NVIDIA Jetson
Infra & tools: Docker, Cloud Run, Cloudflare Workers, GitHub Actions, React, Next.js, Tailwind, Claude Code, Codex

EDUCATION

Keshav Memorial Engineering College — B.Tech, Computer Science & Engineering · CGPA 7.5 / 10 **Expected 2028**